

$\therefore$ By modus ponens, a is even. By Defn. 1, $\exists$ integer K_1 s.t.

$$a = 2K_1. \therefore a^3 = (2K_1)^3 = 8K_1^3 \quad \therefore 2b^3 = 8K_1^3 \Rightarrow b^3 = 4K_1^3 = 2(2K_1^3)$$

By Defn. 1, b^3 is even. By modus ponens, b is even. By Defn. 1, $\exists$ integer K_2 s.t. $b = 2K_2$. $\therefore \frac{a}{b} = \frac{2K_1}{2K_2} = \frac{K_1}{K_2}$. 2 is a common factor of a and b .

This is a contradiction to our assumption that $\frac{a}{b}$ must be in lowest terms.

$\therefore \sqrt[3]{2}$ is irrational $\square$